

PVE — Stage 2: Valuation Engine & Liquidation Waterfall

Independent marks vs self-marks · iterative-conversion waterfall · DLOM · repo ~/dev/pve · 2026-07-05

Waterfall gate 14 tests (hand + Nash oracle + props)	Payoff curve evaluate() = waterfall <1e-9	Recovery n=500 growth 6.08 · margin 4.30 (truth in 95% CI)	Drift detected indep/self = 0.89 (200-book)
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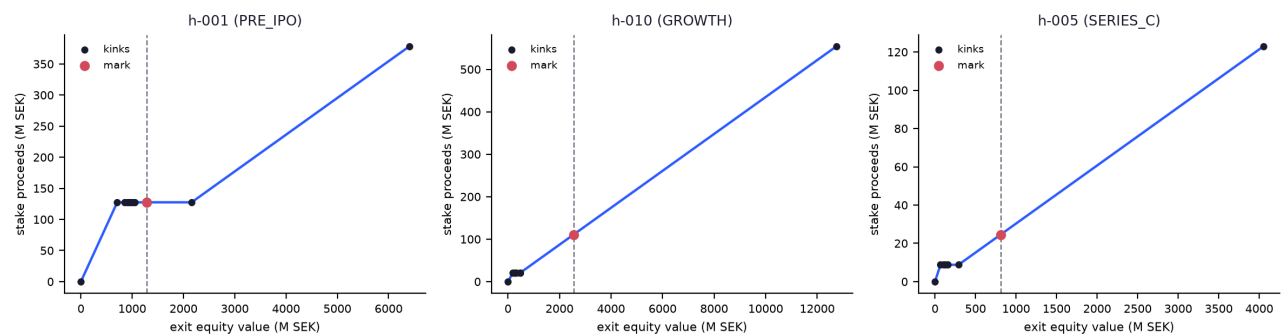
The waterfall (Component D) — the spine

Built and tested first and hardest: a pure single-pass distribution (seniority preference round + exact water-filled residual with participation caps), an iterative conversion fixed-point (each class converts iff it strictly gains), and an exact piecewise-linear payoff curve with kinks located by bisection. Verified by 5 hand-computed known-answer cases, a conversion-interaction case checked against a brute-force Nash-equilibrium oracle, a near-cycle stress case, and property tests (conservation, non-negativity, monotonicity, seniority, determinism) over hypothesis cap tables.

Worked waterfall examples (computed vs. expected)

case	E	computed	expected	ok
1x non-part, E=6 (low)	6	A=6.00, Common=0.00	A=6.00, Common=0.00	PASS
1x non-part, E=30 (converts)	30	A=15.00, Common=15.00	A=15.00, Common=15.00	PASS
1x participating, E=20	20	A=15.00, Common=5.00	A=15.00, Common=5.00	PASS
1x part capped 3x, E=55 (capped)	55	A=30.00, Common=25.00	A=30.00, Common=25.00	PASS
stack, E=90 (senior converts)	90	Senior=26.67, Junior=36.67, Common=26.67	Senior=26.67, Junior=36.67, Common=26.67	PASS

Payoff curves — stake proceeds vs. exit equity (kinks marked, mark in red)

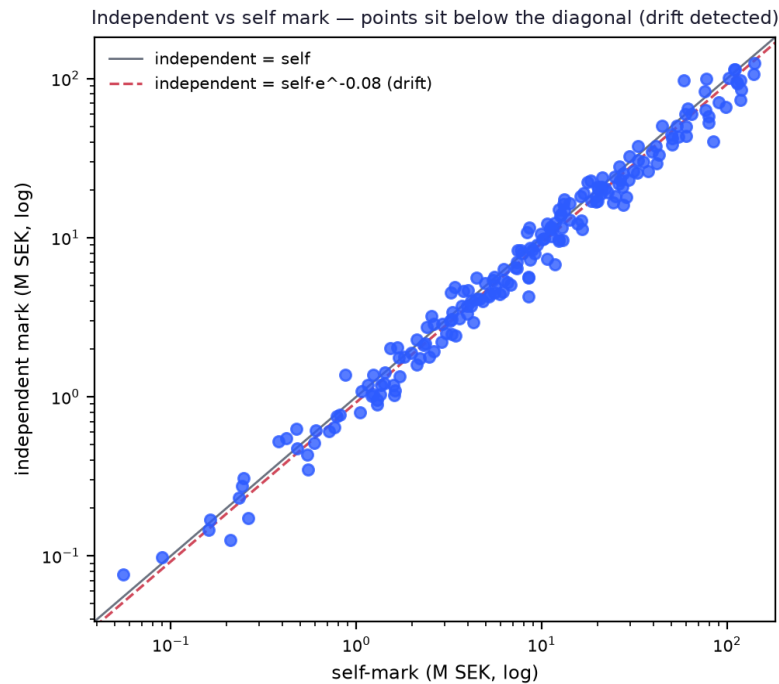


Piecewise-linear in equity value, with corners at preference-exhaustion, participation caps and conversion flips. h-001's Pre-IPO class carries a 1.69x preference (visible as the flat preference plateau before it converts). Stages 3-4 will sample these curves rather than re-deriving.

Demo-book valuation — self vs. independent (sorted by gap)

holding	self-mark	independent	gap	gap%	DLOM	flag
h-006	27.47	16.12	-11.34	-41.3%	16%	OVERMARKED
h-010	113.81	93.04	-20.77	-18.2%	16%	OVERMARKED
h-000	1.21	1.01	-0.20	-16.2%	35%	OVERMARKED
h-005	22.62	19.17	-3.45	-15.2%	22%	OVERMARKED
h-002	0.71	0.60	-0.11	-15.2%	35%	OVERMARKED
h-008	21.20	23.97	+2.76	+13.0%	16%	conservative
h-003	5.61	5.02	-0.59	-10.6%	22%	OVERMARKED
h-004	8.90	8.44	-0.46	-5.2%	22%	
h-001	110.49	114.76	+4.27	+3.9%	10%	
h-009	24.73	24.30	-0.43	-1.7%	10%	
h-007	0.61	0.62	+0.01	+1.7%	35%	
h-011	2.92	2.87	-0.05	-1.6%	25%	
TOTAL	340.27	309.93	-30.34	-8.9%		

Drift detection — the product test



The synthetic self-marks carry a known lognormal optimism drift ($\mu=0.08$, $\sigma=0.22$) over the engine's own fair value. The engine reads it back: aggregate independent/self = 0.886 on a 200-holding book ($\sim 1/E[\text{drift}]=0.90$), and the recovered drift $\mu = 0.091$ matches the planted 0.08. Points sit below the diagonal — the book is systematically over-marked, exactly as planted. (Closed-loop validation; the engine's fairness is established independently by the waterfall known-answers and the $n=500$ multiple recovery.)

Judgment calls

Self-mark recalibration (approved): Stage 0+1 generated $\text{self_mark} = \text{cost_basis} \times \text{drift}$, a naive basis that ignored the cap structure and sat on a different valuation scale than the comps engine, so the $\sim 10\%$ drift was swamped. Recalibrated (valuation layer, provider untouched) to $\text{self_mark} = \text{independent_fair_value} \times \text{drift}$. **DLOM central = empirical** (most directly defensible); Finnerty/Chaffe reported as the range. **Net debt ~ -cash** for the EV $\rightarrow$ equity bridge (no debt data). **None-metric holdings** value via last-round post-money fallback.